In Plain English

CRE Deal Pulse, in plain English

What it is

Deal Pulse is an AI assistant for commercial real estate analysts. It watches the world for them.

When an analyst is working on a deal , say, buying an office building in Midtown Manhattan, they make a bunch of assumptions. They assume the building can charge $74 per square foot in rent. They assume interest rates stay around where they are today. They assume comparable buildings nearby are selling at a certain price. Those assumptions go into a spreadsheet, and the spreadsheet tells them whether the deal is worth doing.

The problem is that the world doesn't sit still. Rates move. A building down the street sells for less than expected. A new lease signs at a lower rent. Any one of those things can quietly break the analyst's spreadsheet , and by the time they notice, they've either missed the warning or wasted weeks pursuing a deal that no longer makes sense.

Deal Pulse watches all of that for them. It knows what the analyst assumed. It checks the real world every day. And when reality stops matching the assumptions, it tells them , in plain English , exactly how much money is on the line and what to do next.

Who it's for

A commercial real estate acquisitions analyst. Usually junior or mid-level. Works at a small-to-midsize real estate firm. Has three or four deals in their pipeline at any given time. Spends their morning tabbing between five different websites , news, government data portals, comparable sales databases, Federal Reserve rate trackers , trying to figure out if anything changed overnight that affects their deals.

Big firms have Bloomberg terminals and entire research departments doing this. Small firms have one tired analyst with twenty browser tabs. Deal Pulse is for the second group.

The user journey, start to finish

Monday morning, 8:45 AM. An analyst named Sarah is working on acquiring a Midtown South office building. Her firm is a week away from submitting a Letter of Intent , the formal "we want to buy this" document. She's already done the underwriting. Her spreadsheet says the deal returns 14% , comfortably above the 12% her firm needs to greenlight.

Before her 9:00 AM stand-up with the partners, she opens Deal Pulse.

She has already loaded her deal into the system once. It knows the address, the square footage, what she's assuming for rent, what cap rate she's using, what her projected NOI is, and what IRR her model spits out. She filled this in on the day she first underwrote the deal. She doesn't fill it in again.

She clicks run. The agent goes to work. It checks three things in parallel:

First, it pulls every open code violation on the actual building from New York City's housing department. Are there fifty open violations she didn't know about? Did three new ones get filed last week? That changes the property's risk profile.

Second, it pulls every comparable sale and lease in the surrounding submarket from the city's property records. Did three nearby buildings just sign leases at $68 per square foot? Sarah is assuming $74. That's a problem.

Third, it pulls the latest interest rates from the Federal Reserve. Did the 10-year Treasury move 25 basis points since she locked her assumptions? That changes her cost of debt, which changes her returns.

Then it does math. Not "ask the AI to do math" , actual deterministic math, the kind that runs the same way every time. It plugs the new observed values into Sarah's underwriting model and calculates: if rent really is $68 and not $74, your NOI drops by $180,000 a year, and your IRR drops from 14% to 11%.

Then it scores. Each finding gets a materiality score from 1 to 5. A score of 1 means "noise, ignore." A score of 5 means "this could kill the deal." The system only surfaces things scored 3 or higher. Sarah doesn't get buried in 47 minor findings , she sees the three things that actually matter.

Then it pauses. This is the most important part. If anything scores a 5, the agent stops. It shows Sarah its reasoning and asks her to confirm before it produces the final briefing. She gets to say "yes, that's a real concern, include it" or "no, you misread that, ignore it." The agent doesn't auto-fire alarms to the partners. A human is always in the loop on the highest-stakes calls.

Finally, it writes the briefing. One page. Top alerts at the top, full change log below. Plain English, with numbers attached.

> *"Market rent in Midtown South dropped to $68/SF this week based on three new lease signings. Your deal assumes $74/SF. At observed rents, NOI falls by $180K and IRR drops from 14% to 11%. Recommend re-running underwriting before LOI."*

Sarah reads it in 30 seconds. She walks into her 9:00 AM stand-up and says, "We need to talk about Midtown South before we send the LOI on Friday." The partners take her seriously because she has the numbers, the sources, and the recommendation already in hand.

She didn't open CoStar. She didn't read three articles. She didn't manually update her model. The agent did it. She made the decision.

Why this matters

The cost of missing a signal is enormous. A $6 per square foot drift in market rent on an 85,000 square foot building is roughly half a million dollars a year in NOI. At a 5.5% cap rate, that's almost ten million dollars in property value. Analysts catch this kind of thing late all the time , not because they're bad at their jobs, but because there's too much to watch and not enough hours.

The cost of false alarms is also real. If you flag every minor signal to the partners, they stop listening. Deal Pulse only surfaces what's actually material, scored against the specific deal the analyst is working on. A 25 basis point rate move matters more on a highly leveraged deal than a lightly leveraged one. The agent knows the difference because it knows the deal.

The math has to be trustworthy. If the analyst is going to walk into a partner meeting and say "our IRR dropped three points," that number has to be right. Deal Pulse computes it deterministically in code, not by asking a language model to do arithmetic. The AI's job is to read the world and explain what it found. The math runs in Python, the same way every time.

Human judgment is preserved on the highest-stakes calls. When something looks like it could kill a deal, the agent stops and asks. It doesn't escalate to leadership on its own. The analyst stays in control.

What it isn't

It's not a deal-finder. It doesn't go scout new acquisitions. It watches deals the analyst is already working on.

It's not a replacement for the analyst's judgment. It surfaces signals and computes impact. The decision , pursue, renegotiate, walk away , is still the analyst's.

It's not a black box. Every flagged signal links back to the source. The analyst can verify, quote, and defend it.

It's not autonomous. A human confirms anything that could kill a deal before it goes to leadership.

Why now, why this team, why this approach

NYC has the best free public real estate data in the country. Violations, sales, ownership, tax assessments , all available, all free, all queryable. Combined with free macro data from the Federal Reserve, you can build something genuinely useful without paying a Bloomberg or CoStar bill that small firms can't afford.

The AI piece is what makes it work. Three years ago, you could have written a script that pulled all this data, but you couldn't have asked it to read three lease comps, score how material they are *to a specific deal*, and write the result up in language a partner would actually read. That's what changed. The model handles the synthesis and the language. The code handles the math and the data.

This week, we ship the first working version. Next week, we upgrade it to a frontier model and add the things only that level of intelligence unlocks: memory across runs, daily diffs that catch what changed since yesterday, and reasoning across signals , *"rates moved and a comp traded; together that means something neither one means alone."*

For a stakeholder, the question to keep in mind is: an analyst at a small firm spends two to three hours a day on this work, catches material signals late, and burns weeks on deals that quietly stopped penciling. Deal Pulse compresses that to thirty seconds and a one-page briefing. The math is right, the sources are linked, the analyst stays in charge.

That's the product.

PRD V2

CRE Deal Pulse , PRD

Project: CRE Deal Pulse Agent

Owner: Joshua Viera

Author: Gamaliel Leguista (v1), updated April 28, 2026

Cohort: Pursuit AI-Native Fellowship, Cycle 3

Demo: Saturday, May 2, 2026

Problem

A CRE acquisitions analyst spends a meaningful chunk of every day refreshing the same handful of inputs to check whether their deal still works. Rates moved, a comp traded wider, a tenant filed something, a violation hit the target property. Any one signal can change the underwriting or kill the deal, but the integration happens in the analyst's head, across multiple tabs and tools. Material changes get missed until they cost real money.

Supporting context:

- Junior analysts spend hours daily checking CoStar, FRED, news feeds, and city portals for changes that affect deals already in pipeline.
- A $6/SF drift in market rent or 25 bps on cap rate can swing IRR by several points; analysts often catch this late.
- Partners ask for "what changed today" updates faster than manual scans can produce.

Opportunity

Build a small agent that watches NYC public data and macro signals against a specific deal's underwriting assumptions, computes the dollar and IRR impact deterministically, and surfaces only what is material. Replace the "five tabs and a spreadsheet" workflow with a one-page briefing the analyst can scan in 30 seconds.

Market opportunity:

- NYC has the strongest free public real estate data in the US (HPD, ACRIS, DOF on Socrata), making this defensible without paid data subscriptions.
- Pursuit fellows and small acquisitions shops are an underserved segment; large funds use Bloomberg/CoStar/Argus stacks, smaller shops use Excel and instinct.

Users & Needs

Primary user: NYC commercial office acquisitions analyst, junior to mid-level, working deals at LOI to under-contract stage. *(Persona locked to office for the staged demo deal , Midtown South, per Elliot's framing example.)*

Secondary users: Their manager or partners receiving escalated alerts.

Needs:

- As an analyst, I need to know when observed market data diverges from my deal assumptions, because I cannot manually scan five sources daily.
- As an analyst, I need the dollar and IRR impact of a change computed for me, in order to decide whether it is material enough to flag to partners.
- As an analyst, I need a human checkpoint before any high-severity escalation, in order to confirm or override the agent's reasoning rather than have it auto-ping leadership.
- As an analyst, I need source links on every flagged signal, in order to verify and quote the source when I escalate.

Proposed Solution

An agent that takes a deal profile (address, BBL, underwriting assumptions) and watches three buckets of signals: property-level (HPD violations on the target asset), market-level (ACRIS recent sales comps), and macro-level (FRED rates including 10-year Treasury and SOFR). When observed values diverge from assumed values, a Python tool computes the NOI and IRR delta deterministically. The LLM narrates the impact in plain English, scores materiality from 1 to 5, and pauses for human review on the highest-severity alerts before producing a markdown briefing.

Top 3 MVP value props:

- **Vitamin:** One place to see every relevant signal on a specific deal, refreshed on demand.
- **Painkiller:** Translates "the 10-year moved 15 bps" into "your IRR drops from 14% to 11%."
- **Steroid:** Produces the demo-grade briefing line: *"Market rent in Midtown South dropped to $68/SF this week. Your deal assumes $74. NOI falls by $180K. Recommend re-running underwriting before LOI."*

Goals & Non-Goals

Goals:

- Ship v1 by Tuesday EOD with three signal tools and one deterministic math tool wired into a working agent loop.
- Land a 3-minute Saturday demo on a single pre-staged deal, with the killer-quote briefing as the centerpiece.
- Build a clear v1 → v2 narrative so the Claude upgrade week shows visible improvement.

Non-goals:

- Not building daily diffs or memory across runs in v1 (planned for v2).
- Not building multi-step reasoning across signal buckets in v1 (planned for v2).
- Not building tenant credit watch, DOB permits, or tax delinquency tools in v1.
- Not building any UI beyond a generated markdown briefing.
- Not supporting multiple deals; v1 ships with a single staged demo deal.
- Not running fully autonomously; the human checkpoint is a feature, not a limitation.

Team & Roles

Person	Role	Owns
Joshua	Engineering lead, integration owner	Repo, scaffold, agent loop, `compute_underwriting_delta`, final audit
Pedro	Tool 2 , Market signals	`tools/market_signals.py` (ACRIS), live demo segment
Elliot	Tool 3 , Macro signals + deck clicker	`tools/macro_signals.py` (FRED), slide advancement during demo
Kevin	Slides + QA	Demo deck, integration testing alongside Gamaliel
Gamaliel	Merge + QA	PR merges, integration testing, presenting alongside Pedro

Presenting on Saturday: Gamaliel and Pedro lead. Elliot clicks slides. Kevin and Joshua handle live-run support.

Success Metrics

Goal	Signal	Metric	Target
Demo lands	Stakeholder leans forward at the briefing line	Briefing contains correct dollar and IRR impact	100% accurate on staged deal
Tool reliability	Agent completes a run without hand-holding	Successful tool calls per run	4/4 tools called end-to-end
Team coverage	Every member can speak to the work	Members able to present a component live	5/5 team members

v1 → v2 story	Frontier model shows visible lift	Capabilities only Claude unlocks (planner, memory, multi-signal reasoning)	3 demonstrable in v2 demo

---

Deal Profile Schema

The deal profile is the single shared input every tool and the math function reads from. Locked here so Pedro, Elliot, and the underwriting delta function don't reconcile field names mid-integration.

```
{
  "deal_id": "midtown-south-office-001",
  "property": {
    "address": "150 W 30th St, New York, NY 10001",
    "bbl": "1008060001",
    "borough": "1",
    "submarket": "Midtown South",
    "asset_class": "office",
    "square_footage": 85000,
    "top_tenants": ["Tenant A", "Tenant B", "Tenant C"]
  },
  "deal_stage": "LOI",
  "underwriting": {
    "market_rent_psf": 74.00,
    "in_place_rent_psf": 68.00,
    "vacancy_rate": 0.08,
    "going_in_cap": 0.055,
    "exit_cap": 0.060,
    "rent_growth": 0.03,
    "hold_period_years": 5,
    "noi": 5_200_000,
    "irr": 0.14
  },
  "assumptions_locked_at": "2026-04-25"
}
```

Field contract:

- All rates and percentages are decimals (0.14, not "14%").
- Dollar values are integers or floats, no formatting.
- `bbl` is a 10-digit string, validated by `tools/violations.py`.
- `borough` is a single character "1"–"5" matching ACRIS conventions.

Requirements

Legend: [P0] = MVP for v1 demo · [P1] = polish · [P2] = nice-to-have

Use Case 1: Analyst configures a deal and runs the agent

Setting up the deal profile

- [P0] User can pre-stage a deal profile in JSON matching the schema above.
- [P0] Agent loads the deal profile at runtime.
- [P1] User can edit assumptions in the JSON and re-run without code changes.
- [P2] User can manage multiple deal profiles in a single config (v2).

Running the agent loop

- [P0] Agent calls property signals tool (HPD violations on target BBL).
- [P0] Agent calls market signals tool (ACRIS recent sales for the submarket and asset class).
- [P0] Agent calls macro signals tool (FRED 10-year Treasury and SOFR).
- [P0] Agent calls deterministic underwriting delta tool with assumed values and observed values.
- [P1] Agent runs the three signal tools in parallel to keep latency reasonable.
- [P2] Agent decides which tools are worth calling for this specific deal rather than running all of them (v2 planner).

Scoring materiality and reviewing alerts

- [P0] LLM assigns severity 1–5 to each signal based on the computed delta, not on the raw signal.
- [P0] Briefing surfaces only signals with severity ≥ 3.
- [P0] If any signal scores severity 5, agent pauses and shows its reasoning before continuing.
- [P1] User can confirm or override at the checkpoint, and the override is logged.
- [P2] Agent learns from overrides across runs to recalibrate severity scoring (v2 memory).

Briefing output

- [P0] Agent produces a markdown briefing with top alerts at the top and the full change log below.
- [P0] Each high-severity alert uses the demo-style language: observed value, assumed value, computed dollar/IRR impact, recommendation.
- [P1] Briefing includes source links for every cited signal.
- [P2] Briefing delivered via email or Slack webhook (v2).

Use Case 2: Saturday demo day

- [P0] Live run on the staged deal produces correct numbers.
- [P0] Each presenting team member can speak to at least one component.
- [P0] Demo arc covers: deal profile (15s), live agent run with visible tool calls (45s), briefing output (45s), human checkpoint moment (30s), v2 setup (15s).
- [P1] Backup pre-recorded run is ready in case the live API call fails.
- [P2] Slide framing the v1 → v2 story so the upgrade week has clear stakes.

Open Questions

- ~~Persona lock: multifamily or office?~~ **Resolved: office, Midtown South.**
- Is per-property watch list with daily diffs a v2 stretch goal or a v3 roadmap slide?
- Does the briefing go to console only for v1, or do we want a Slack delivery for the demo? *(Leaning console for v1; Slack is a v2 polish item.)*
- Tenant credit (SEC EDGAR) for v2: does it strengthen the product, or does it pull us back toward the broader distressed-lead-gen idea we de-scoped?

Appendix

- **Repo:** https://github.com/JoshuaViera/cre-distress-agent
- **Designs:** TBD
- **Other resources:**
    - Elliot's framing memo (Market / Property / Macro signal buckets)
    - Pedro's Deal Pulse architecture sketch (deal profile in, daily scan, human checkpoint)
    - HPD violations tool (shipped Apr 27, slots into Property signals bucket)

Tech/Roadmap

CRE Deal Pulse — Technical Roadmap

Executive summary

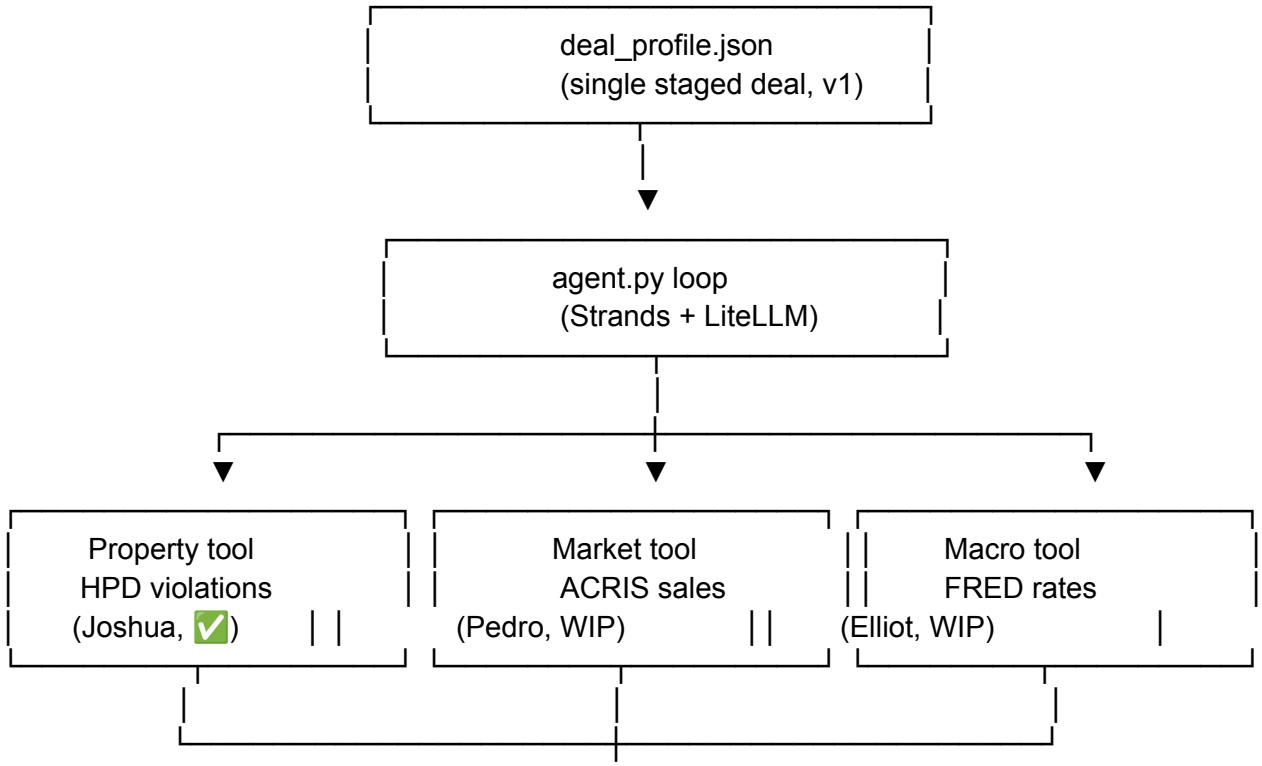
Deal Pulse ships in two waves. Wave one — through Saturday May 2 — is a working agent on a free model that hits Elliot's killer-quote demo on a single staged deal. Wave two — through Friday May 9 — upgrades the brain to Claude and adds the capabilities that brain unlocks: memory across runs, daily diffs, and multi-signal reasoning.

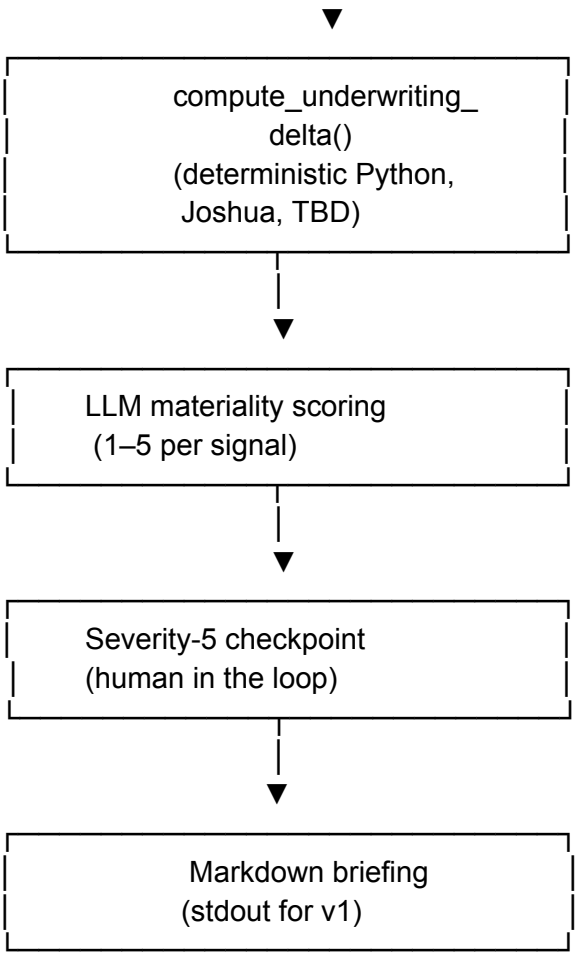
The architecture stays the same across both waves. The same code that runs on Hy3 Preview today runs on Claude next week. The point of the v1 → v2 story is to show stakeholders exactly what changes when you swap the model — and what doesn't.

Wave 1 outcome (Saturday): A 3-minute live demo on a staged Midtown South office deal. Three signal tools fire, deterministic math runs, the briefing lands the killer line, the human checkpoint pauses for confirmation. Stakeholders lean forward. Anthropic API keys unlock.

Wave 2 outcome (Week 2): The same demo, plus three things only a frontier model makes credible: the agent remembers prior alerts and recalibrates, it tells the analyst what changed since yesterday, and it reasons across signals — *"rates moved and a comp traded; together that means something neither one means alone."*

Architecture at a glance





The model swap point: the box labeled *"LLM materiality scoring"* runs on Hy3 Preview in Wave 1 and on Claude in Wave 2. Everything else is identical.

Wave 1 — through Saturday May 2

Outcome

A live, working demo on a staged deal that produces the killer-quote briefing, pauses on a severity-5 alert, and finishes inside three minutes.

What's already shipped (as of April 28)

- Repo scaffold, dependencies, environment configuration
- `test_model.py` — confirms Hy3 round-trips through OpenRouter
- `tools/violations.py` — Property signals tool, full contract, error and empty-case handling
- `agent.py` — minimal loop wiring Hy3 + the violations tool with a CRE system prompt
- Demo BBL `2026140035` produces a clean analyst-style assessment end-to-end

What ships this week

Tool 2 — Market signals (ACRIS sales)

Owner: Pedro · **Branch:** `feat/tool-market-signals` · **Status:** in progress

- `tools/market_signals.py` with contract `get_market_signals(borough, days_back=90, min_sale_price=1_000_000)` returning sale_count, median_price, sample_sales, market_signal
- Hits ACRIS Master + Legals joined by document_id, filtered to DEED/DEEDO doc types
- Acceptance: Manhattan and Brooklyn both return real data; bad borough returns error envelope; empty result returns `market_signal: "no_data"`
- Wired into `agent.py` with `@tool` decorator following the `get_property_distress_signals` pattern

Tool 3 — Macro signals (FRED rates)

Owner: Elliot · **Branch:** `feat/tool-macro-signals` · **Status:** in progress

- `tools/macro_signals.py` with contract `get_macro_signals(days_back=30)` returning current and prior values for 10Y Treasury (DGS10) and SOFR, plus signed bps_change and a top-level macro_signal
- Acceptance: real numbers for both series; missing FRED_API_KEY returns error envelope; bps_change math eyeballed against two known FRED dates
- Sign convention: positive bps_change = rates rose (bad for the deal), negative = rates fell (good for the deal). Narration must preserve direction.

Tool 4 — Deterministic underwriting math

Owner: Joshua · **Status:** not started — **highest priority unblocked item**

- `tools/underwriting.py` with `compute_underwriting_delta(deal_profile, observed_rent_psf=None, observed_cap=None, observed_rate_bps=None) → dict`
- Returns: `{noi_delta_dollars, irr_delta_pct, narrative_inputs}` — pure Python, no LLM, deterministic
- Acceptance: feeding Elliot's killer-quote scenario ($74 → $68 rent on the staged deal) produces ~$180K NOI delta and ~3pt IRR drop. If it doesn't, the staged deal's inputs get tuned until it does.
- This blocks the briefing's killer line — must land before Pedro's and Elliot's tools are integrated end-to-end

The staged deal profile

Owner: Joshua + Gamaliel · **Status:** schema locked in PRD, instance not yet created

- `deals/midtown-south-office-001.json` matching the schema in the PRD
- Tuned so the math function produces the demo numbers
- Loaded by `agent.py` at startup; no other deals supported in v1

Human checkpoint

Owner: Joshua · **Status:** not started

- After the LLM scores materiality, if any signal scores 5, the agent prints its reasoning and waits for `[y/n]` on stdin before producing the final briefing
- Override (n) is logged to `runs/{timestamp}.log` for audit
- Required by the cohort brief — non-negotiable

Briefing output

Owner: Joshua · **Status:** not started

- Markdown, top alerts (severity ≥ 3) above the fold, full change log below
- Each alert: observed value, assumed value, computed dollar/IRR impact, recommendation, source link
- v1 destination: stdout. Slack and email are v2.

Day-by-day

Day	Date	Goal	Owner
Tue	Apr 28	Tools 2 + 3 PRs land. Math function written and tested against killer-quote scenario. Staged deal profile created. Human checkpoint scaffolded.	Pedro, Elliot, Joshua
Wed	Apr 29	Cross-cohort hackathon — different mode, no Deal Pulse work	—
Thu	Apr 30	Integration testing. All four tools fire end-to-end. Briefing formatter complete. Backup recording captured.	Kevin, Gamaliel
Fri	May 1	Demo rehearsal with timer. Failure-mode rehearsal (run the "backup" pivot at least once). README and ADR finalized.	Full team
Sat	May 2	**Demo Day v1.0**	Gamaliel + Pedro present, Kevin/Elliot drive laptop, Joshua on standby

Wave 1 success criteria

- Live run on the staged deal completes in under 90 seconds
- Briefing contains the killer line with correct dollar and IRR figures
- Severity-5 checkpoint pauses and resumes cleanly
- All five team members can speak to at least one component
- Backup recording exists and is rehearsed

Risks and mitigations

Risk	Likelihood	Mitigation
ACRIS or FRED rate-limits during the live demo	Medium	Backup pre-recorded run, rehearsed pivot, hotspot ready

Hy3 leaks reasoning text into the briefing	Medium	`max_tokens=8192` already set; system prompt instructs clean output; LiteLLM logging suppressed
`compute_underwriting_delta` math doesn't produce demo numbers	High if not done early	Build the math function first, tune the staged deal's inputs to match
One of Pedro or Elliot's PRs slips past Tuesday	Medium	Wednesday is hackathon, Thursday is buffer — but if either slips to Friday, the integration window collapses
BBL validation rejects the staged office deal's BBL	Low	Pre-test the staged deal's BBL through the violations tool before Saturday

---

Wave 2 — through Friday May 9 (Claude upgrade week)

Outcome

The same demo, on Claude, with three new capabilities visible side-by-side with v1 output. The pitch: *"this is what changes when you swap the model."*

Conditional on

Anthropic API keys unlocking after the Saturday presentation. If they don't, Wave 2 either runs on Hy3 with the same new features (weaker story) or pauses pending key access.

What gets built

Model swap

Owner: Joshua · **Effort:** half a day

- Replace the `LiteLLMModel(model_id="openrouter/tencent/hy3-preview:free", ...)` config with the Anthropic Claude config
- Same Strands wiring, same tool definitions, same system prompt initially
- Smoke test: rerun the Saturday demo on Claude and confirm the briefing reads cleaner (no leaked reasoning, sharper materiality scoring)

Capability 1 — Daily snapshot + diff

Owner: Joshua + Pedro · **Effort:** 2 days

- Each agent run writes a snapshot to `snapshots/{deal_id}/{YYYY-MM-DD}.json` containing the raw tool outputs
- On the next run, the agent loads the most recent snapshot, computes a structured diff against today's tool outputs, and feeds the *diff* to the materiality scorer instead of the raw values
- Briefing changes from *"market rent is $68"* to *"market rent dropped from $74 to $68 since yesterday"*

- This is the capability that makes Deal Pulse feel like a daily tool instead of a one-shot scanner

Capability 2 — Multi-signal reasoning

Owner: Joshua + Elliot · **Effort:** 1 day

- New synthesis step after individual scoring: pass all signals together to Claude with a prompt that asks *"are any of these correlated, compounding, or canceling each other out?"*
- Output: a "compound finding" block at the top of the briefing when correlation is detected
- Demo line: *"The 10-year moved 25 bps wider AND a comparable just traded at a 6.2 cap. Together this implies the market is repricing — your 5.5 going-in cap looks light."*
- This is the capability Hy3 cannot credibly do — Claude's reasoning depth is the entire point

Capability 3 — Memory across runs

Owner: Joshua + Kevin · **Effort:** 1.5 days

- Each human-checkpoint decision (confirm/override) gets logged with the alert's reasoning
- On subsequent runs, the materiality scorer is shown the recent override history for similar signals as context
- Effect: if the analyst dismissed three rate alerts in a row as "not material to this deal," the agent stops scoring rate moves as 5s on this deal. It learns this analyst's threshold for this deal.
- Demo line: *"Last week you dismissed two similar rate-move alerts as immaterial to this deal. I've recalibrated and scored this one a 3 instead of a 5."*

Capability 4 — Real planner (stretch)

Owner: Joshua · **Effort:** 1 day · **Status:** stretch, only if the above three land cleanly

- Instead of running all three tools every time, Claude decides which tools are worth calling based on the deal profile and the most recent diffs
- Example: if rates haven't moved in five days, skip the FRED call this run
- Saves API quota, but more importantly: shows the agent doing actual planning, not just executing a fixed sequence

Day-by-day (Wave 2)

Day	Date	Goal
Sun	May 3	Buffer / rest
Mon	May 4	Anthropic key swap. Side-by-side smoke test (same staged deal, both models).
Tue	May 5	Snapshot + diff infrastructure. First run with diff-based scoring.
Wed	May 6	Multi-signal reasoning step. First compound finding in a briefing.

Thu	May 7	Memory across runs. Override logging plumbed into materiality scorer context.
Fri	May 8	Polish, side-by-side demo recording (Hy3 vs Claude on identical input), v2 README
Sat	May 9	Wave 2 cohort checkpoint or stakeholder follow-up

Wave 2 success criteria

- Side-by-side recording shows Hy3 v1 output and Claude v2 output on the same staged deal, same data
- All three v2 capabilities (diff, multi-signal, memory) are demonstrable in a single run
- The v2 briefing is visibly sharper than v1 — different in kind, not just degree
- A clear narrative slide exists for "what changed and why it matters"

Out of scope for both waves

These are deliberately deferred. If a stakeholder asks "why don't you have X," the answer is *"v3 — here's the order we'd build it."*

- Multiple deals in a single config
- DOB permits, DOF tax assessments, tenant credit (SEC EDGAR)
- Slack and email delivery
- Web UI
- Auth, multi-tenancy, billing
- Census or BLS submarket signals
- Caching layer for NYC Open Data rate limits (will become necessary if usage grows; not a v1/v2 problem)

Decisions log

Date	Decision	Reason
Apr 27	Pivot from "distressed multifamily acquisitions" to "Deal Pulse for active deals"	Pedro's design doc + Elliot's framing aligned to monitoring an analyst's existing pipeline, not hunting new deals. HPD tool survives the pivot.
Apr 27	Math runs in Python, LLM narrates	"We compute deterministically, the agent narrates" is a stronger story than "the LLM did some math, hopefully right."
Apr 28	Persona locked to office (Midtown South), not multifamily	Elliot's killer-quote example is office; the staged deal commits to one, math differs by asset class.

Apr 28	Briefing destination is stdout for v1, Slack/email is v2	Cuts integration work in week one; doesn't weaken the demo.
Apr 28	Human checkpoint blocks at severity 5, not lower	Required by cohort brief; lower thresholds would create alert fatigue in the demo.

Open questions

- Does Wave 2 stretch goal (real planner) ship if capabilities 1–3 take longer than estimated, or does it get cut cleanly?
- Is there a stakeholder beyond Anthropic and Pursuit instructors we should be designing the v2 demo for? (E.g., a real CRE analyst willing to validate the briefing language.)
- After May 9, does this become a fellowship portfolio piece, an open-source tool, or something the team continues building? Decision before May 9 affects how the v2 README and license get framed.

Script

CRE Deal Pulse — Demo Script

Saturday, May 2, 2026 · 3-minute stakeholder demo

Cast & roles

Person	Role on stage
Gamaliel	Opens. Sets up the problem and the deal.
Pedro	Closes. Lands the takeaway and the v2 setup.
Kevin or Elliot	Drives the laptop. Runs the live agent. Advances slides.
Joshua	On standby for live-run support.

Total runtime: 3:00. Budget every second.

Pre-demo checklist (do this 10 minutes before going live)

- Laptop on Wi-Fi, hotspot ready as backup
- `.env` has `OPENROUTER_API_KEY` and `FRED_API_KEY` filled in
- Run `python test_model.py` once — confirm Hy3 round-trips
- Open the staged deal JSON in a separate window, ready to show
- Backup recording loaded in a browser tab, ready to play if live run fails
- Terminal font size bumped to 18pt minimum so the room can read it
- LiteLLM logging suppressed (already in `agent.py`, just confirming)
- Slack and email notifications muted on the demo machine

Segment 1 — The hook (0:00–0:15) · GAMALIEL

> *Walks on. Slide 1 up: "CRE Deal Pulse" + the killer-quote line.*

Gamaliel:

> "A commercial real estate analyst is one week away from sending a Letter of Intent on a Midtown South office building. Her model says the deal returns 14%. This morning, three comparable

buildings signed leases at six dollars per square foot below what she assumed. She doesn't know yet. By the time she finds out, she's three weeks deeper into a deal that no longer pencils."

(One beat. Looks at the room.)

"We built the agent that tells her this morning."

Segment 2 — The deal profile (0:15–0:30) · GAMALIEL

Kevin/Elliot switches screen to the staged deal JSON. Highlight the underwriting block.

Gamaliel:

"This is her deal. Eighty-five thousand square feet in Midtown South. She's underwriting at seventy-four dollars per square foot market rent, a five-and-a-half cap going in, and a fourteen percent IRR. She loaded these assumptions into Deal Pulse on the day she first underwrote the deal. She doesn't fill them in again."

(Beat.)

"Watch what happens when she clicks run."

Segment 3 — The live run (0:30–1:15) · KEVIN or ELLIOT drives, GAMALIEL narrates

Switch to terminal. Run `python agent.py`. The agent calls tools in sequence — visible on screen.

Gamaliel narrates as tools fire:

"First, it pulls every open code violation on the actual building from New York City's housing department."

(HPD tool returns.)

"Then comparable sales in the submarket from city property records."

(ACRIS tool returns.)

"Then the latest interest rates from the Federal Reserve."

(FRED tool returns.)

"Three sources, no logins, no paid subscriptions. All public data. Now watch the math."

(`compute_underwriting_delta` runs. Numbers appear.)

"It just took her assumed rent of seventy-four, the observed rent of sixty-eight, and computed the impact on NOI and IRR — in Python. Not the language model. The math runs the same way every time."

Segment 4 — The briefing and the human checkpoint (1:15–2:30) · PEDRO takes over

Briefing renders to screen. Severity-5 alert triggers checkpoint.

Pedro:

"Here's what the agent produced. One page. Top alerts at the top, full change log below."

(Reads the killer line aloud, slowly.)

"Market rent in Midtown South dropped to sixty-eight dollars per square foot this week based on three new lease signings. Your deal assumes seventy-four. At observed rents, NOI falls by one hundred and eighty thousand dollars and IRR drops from fourteen percent to eleven percent. Recommend re-running underwriting before LOI."

(Beat. Lets it land.)

"That is the entire workflow. Five tabs and a spreadsheet, replaced with one paragraph she reads in thirty seconds."

(Pause. Points to the checkpoint prompt on screen.)

"And before this briefing got finalized, the agent stopped. This finding scored a five — potentially deal-killing. So it surfaced its reasoning and asked the analyst to confirm before producing the final output. It does not auto-escalate to the partners. A human is always in the loop on the highest-stakes calls."

(Kevin or Elliot clicks "confirm." Briefing finalizes.)

"She walks into her nine AM stand-up with the numbers, the sources, and the recommendation already in hand."

Segment 5 — The v2 setup and close (2:30–3:00) · PEDRO

Slide: v1 → v2 capabilities side by side.

Pedro:

"What you just saw is version one, running on a free model, shipped this week. Version two is where this gets sharp."

(Beat.)

"Next week, the same agent runs on Claude. It gains memory across runs — it remembers which alerts the analyst confirmed and which she dismissed, and it recalibrates. It runs daily diffs — it tells her what changed since yesterday, not just what's true today. And it reasons across signals — rates moved and a comp traded, and together that means something neither one means alone."

(One beat. Looks at the room.)

"Small real estate firms don't have Bloomberg terminals or research departments. They have one tired analyst with twenty browser tabs. We built the agent for the second group."

(Final beat.)

"Thank you."

Backup plan — if the live run fails

Trigger conditions: API timeout, rate limit, unexpected exception, agent stalls more than 10 seconds.

Who calls it: Joshua, from offstage. Single word: *"backup."*

What happens:

1. Kevin/Elliot switches to the pre-recorded run (browser tab, already cued).
2. Gamaliel pivots: *"We pre-recorded a clean run earlier — same deal, same data — so you can see the full output."*
3. Pedro picks up at Segment 4 as scripted, reading from the recorded briefing.
4. Do not apologize for the backup. Do not explain why. Move forward.

The recording exists for a reason. Use it without breaking stride.

Timing reference

Segment	Owner	Start	End	Length
1. Hook	Gamaliel	0:00	0:15	0:15
2. Deal profile	Gamaliel	0:15	0:30	0:15
3. Live run	Kevin/Elliot + Gamaliel	0:30	1:15	0:45
4. Briefing + checkpoint	Pedro	1:15	2:30	1:15
5. v2 + close	Pedro	2:30	3:00	0:30

Where time will get eaten: Segment 3 (tool calls take longer than expected). Build the cushion there. If tools run fast, Gamaliel slows the narration. If they run slow, he keeps narrating over the spinner — never let dead air sit.

Lines that must land word-for-word

These three are non-negotiable. Practice them out loud:

1. **The opening:** *"We built the agent that tells her this morning."*
2. **The killer line:** *"Market rent in Midtown South dropped to sixty-eight dollars per square foot this week… NOI falls by one hundred and eighty thousand dollars and IRR drops from fourteen percent to eleven percent. Recommend re-running underwriting before LOI."*
3. **The close:** *"Small real estate firms don't have Bloomberg terminals or research departments. They have one tired analyst with twenty browser tabs. We built the agent for the second group."*

Everything else can flex. These three carry the demo.